# RAUNAK MONDAL

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## EDUCATION

**Carnegie Mellon University, School of Computer Science** Aug 2025 – May 2028 (expected)
*B.S. Computer Science, Machine Learning concentration · Pittsburgh, PA*

**Honors:** 1st (Prelim), CMU March Madness ML · 5th, CMU AI Safety Estimathon · National Merit Finalist · HMMT 21st (Guts) · ARML Top 50 · Stanford SUMaC
**Coursework:** Computer Systems (15-213), Parallel & Sequential Data Structures and Algorithms (15-210), Data Structures & Algorithms (15-122), Functional Programming (15-150), Computer Security (15-330), Probability Theory for Computer Scientists (36-218), Concepts of AI (07-180)

## SELECTED PROJECT

**Image Memorability with a Calibrated Trust Layer** Oct 2026
*independent project · PyTorch, CLIP ViT, Modal · github.com/raunak2007/lamem-memorability*

- Frozen CLIP ViT-B/32 features + ridge predict LaMem memorability at Spearman 0.699 (10k test), reaching the human split-half consistency ceiling ($\rho \approx 0.68$), over a ResNet18 baseline $0.619 \pm 0.009$ and a 0.097 brightness floor (fine-tuned MemNet 0.64). Split-conformal calibration over a heteroscedastic variance head gives near-exact coverage (0.905 at nominal 0.90); selective prediction cuts confident-slice MAE 0.069 $\rightarrow$ 0.058, and OOD distance routes out-of-domain ad creative at AUROC 0.93 where the model's own variance cannot. Reframed as creative A/B selection, the model picks the more memorable of two images 74% of the time, and gating on the prediction margin holds 87% on the confident half it auto-decides while deferring the close calls.

## EXPERIENCE

**Twinly · Software Engineering Intern** May 2026 – Present
*AI desktop agent startup (macOS and Windows) · C#/.NET 9, Electron + React, Avalonia · 30+ PRs, 289 commits*

- Shipped the plugin SDK: plugins declare tools by manifest with a JSON Schema and scoped permissions, the registry enforces at load that each tool's permissions are a subset of the manifest's, and an absent consent resolver defaults to an empty grant, so forgetting to wire consent blocks side-effect tools rather than opening them.
- Led a defensive security audit across 7 attack surfaces, authored a 24-finding severity-ranked report, and landed fixes across 12 files in C#, TypeScript, Python and PowerShell, including bearer-token auth on an unauthenticated sidecar.
- Built PromptLab, the prompt A/B harness, so prompt changes ship on evidence: FNV-1a 64-bit bucketing (`string.GetHashCode()` is randomized per process in .NET Core and would reshuffle every user on restart) with a two-sided pooled two-proportion z-test for rollup.

**ScottyLabs × CMU Graduate Student Assembly · Backend & Database Developer** Oct 2025 – Present
*Production platform · PostgreSQL/Supabase, TypeScript, React · 240 PRs, 512 commits*

- Own the database layer of a production review platform: 85 RPC functions over 57 migrations and ∼19,000 lines of SQL, on a two-schema design where raw submissions land in a private admin schema nothing public reads and every public path returns through a SECURITY DEFINER RPC that aggregates and authorizes first.
- Found and fixed two access-control defects in a self-directed audit (an IDOR in an RPC that trusted a caller-supplied user ID, a missing role check on a publish path) and two projection bugs from user reports.

**Smartbite AI · Founding Engineer** Jan 2025 – Present

- Built the core product from scratch; contributed to the CES 2026 live demo: LLM agent workflows pairing a recommender with nutritional constraint solving.

## RESEARCH

**Do LLM Agents Run Controlled Experiments? (CRN-Bench)** Jan 2026 – Present
*advised by Prof. Amarnath Gupta, UC San Diego*

- Benchmark for whether an agent probing a stochastic simulator holds the seed fixed across single-parameter changes or reads noise as signal. Scripted baselines bound the range before any model is evaluated: drawing fresh seeds confabulates 0 of 15 findings at a 50-call budget and 6 of 15 at 200, because argmax-then-t-test selection reaches significance without becoming real. Motivating case is a 21-parameter simulator I built over the OpenAlex citation graph, where screening found 15 parameters inert, one surfaced to users as a live control.

## OPEN SOURCE

- Correctness fixes merged into major systems, each a quietly wrong output caught with a failing test: **nautechsystems/nautilus_trader** (28k stars, Rust; #4934, a Sortino NaN-guard), **ccxt** (44k stars; #30288, a transpiler regex corrupting two shipped WebSocket subscriptions), **dlt** (#4542, a schema-hash replay on a partial migration, generalized across all destinations), and **OpenAPITools/openapi-generator** (27k stars; three Kotlin fixes in 7.25.0).
- Diagnoses the maintainers shipped in **web-infra-dev/rspack** (ByteDance's Rust bundler; #14862, #14860) and nautilus_trader (#4936, folded into the IB adapter refactor #5041); plus open PRs to **bytedance/bolt** (C++ Spark) and **paradedb** (#6480, approved).

## PUBLICATIONS

Mondal, R. et al. "Which Sequence-Based Neural Network Architecture Yields Optimal Performance in Classification of Autism-Related Behavior." AMIA 2024 (first author). · Mondal, R. et al. "Are Robots Ready to Deliver Autism Inclusion?" ACM CHI 2024 (co-author).

## TECHNICAL SKILLS

**Languages:** Python, C, C#, Rust, Java, TypeScript/JavaScript, SQL **Frameworks:** .NET 9, React, Electron, Node.js, Flask
**ML & Vision:** PyTorch, scikit-learn, NumPy · CLIP/ViT, conformal prediction, selective prediction, uncertainty quantification, OOD detection · hypothesis testing, bootstrap CIs
**Data & Infrastructure:** PostgreSQL (PL/pgSQL, RLS), Supabase, Neo4j, Modal, Docker, Git, Linux